Recursion, bijections and double counting

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L4-CO2 Level 4 Olympiad · Combinatorics

Three counting methods: count one set two ways, match it with something easier, or build it from smaller cases, with the rigour a BMO proof needs.
By the end you can set up and justify recursions, using states when one sequence is not enough; construct bijections and prove them by writing down the inverse; double count incidences and pairs to find exact answers and to prove bounds.
Prerequisites: Stars and bars, inclusion-exclusion and bijections (L3-CO1), Expected value and probability tools (L3-CO2). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Twelve students each belong to exactly three clubs, and every club has exactly five members. How many clubs are there?

Count the pairs (student, club) with the student in the club. Counting by students gives $12 \times 3 = 36$. Counting by clubs gives $5c$ if there are c clubs. So $5c = 36$, which is impossible, and no such arrangement exists.

This unit covers three counting methods: counting one set in two ways (double counting), matching it with a set that is easier to count (bijections), and building it from smaller cases (recursion). Most AIME counting questions, and many BMO combinatorics problems, use one of them.

1.1 Double counting

Double counting.
Let A and B be finite sets and R a set of pairs (a, b) with $a \in A$ and $b \in B$. If $r(a)$ is the number of pairs in R containing a, and $c(b)$ the number containing b, then

$$\sum_{a \in A} r(a) = |R| = \sum_{b \in B} c(b).$$

Both sides count R, sorted by first entry or by second entry. Two consequences you should know:

- **Handshake lemma.** In any group, the sum over people of the number of friends each has equals twice the number of friendships, since each friendship is counted from both ends. In particular the number of people with an odd number of friends is even.
- **Committee identities.** Count committees of k people chosen from n, with a chair: choose the committee and then its chair, or the chair and then the other $k-1$ members. So $k\binom{n}{k} = n\binom{n-1}{k-1}$.

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The main decision is what to count. Usually it is incidences (a person on a committee) or pairs (two people on the same committee).

Example 1.
Ten people form some committees of four. Every two of the ten people serve together on exactly two committees. How many committees are there, and on how many committees does each person serve?

Solution.
Counting incidences does not work straight away, since we know neither the number of committees nor how many each person is on. The information we are given is about pairs, so count pairs.
Count the pairs (two people, committee) where both people are on the committee. By pairs of people: there are $\binom{10}{2} = 45$ of them, each on two committees, giving 90. By committees: each committee contains $\binom{4}{2} = 6$ pairs of people, giving $6c$ if there are c committees. So $c = 15$.
Now fix a person P and count the pairs (another person Q, committee containing both P and Q). By Q: $9 \times 2 = 18$. By committee: each of the r committees containing P contains 3 other people, giving $3r$. So $r = 6$, the same for everyone. As a check, the incidences come to $15 \times 4 = 60 = 10 \times 6$.

Such a system of committees does exist, but the counting determines the numbers without a construction.

1.2 Bijections

Bijection principle.
If $f : A \to B$ has an inverse, that is, a map $g : B \to A$ with $g(f(a)) = a$ and $f(g(b)) = b$ for all $a \in A$ and $b \in B$, then $|A| = |B|$.

In practice you describe the map, describe how to undo it, and check both directions; that is the proof. The bijections you should know by heart:

- subsets of $\{1, \ldots, n\}$ and binary strings of length n, so there are 2^n of each;
- paths of unit steps right and up from $(0,0)$ to (m,n), and words with m letters R and n letters U, so there are $\binom{m+n}{m}$ of each;
- stars and bars, from L3-CO1: solutions of $x_1 + \cdots + x_k = n$ in non-negative integers and rows of n stars and $k-1$ bars.

Example 2.
A row in a cinema has 20 seats. In how many ways can 5 people choose seats (we only care which seats are taken) so that between any two of them there are at least two empty seats?

Solution.
(Shifting out forced gaps.) Casework on the sizes of the gaps is long. Write the chosen seat numbers as $s_1 < s_2 < \cdots < s_5$; the condition is $s_{i+1} - s_i \geq 3$. We remove the empty seats that are forced: set

$$t_i = s_i - 2(i-1),$$

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so $t_1 = s_1$, $t_2 = s_2 - 2$, and so on up to $t_5 = s_5 - 8$.
Then $t_{i+1} - t_i = s_{i+1} - s_i - 2 \geq 1$, hence the t_i increase, with $t_1 \geq 1$ and $t_5 \leq 20 - 8 = 12$. So $\{t_1, \ldots, t_5\}$ is a 5-element subset of $\{1, \ldots, 12\}$. Conversely, any 5-element subset $t_1 < \cdots < t_5$ of $\{1, \ldots, 12\}$ gives $s_i = t_i + 2(i-1)$, which satisfies $s_{i+1} - s_i = t_{i+1} - t_i + 2 \geq 3$ and $s_5 \leq 20$. The two maps undo each other, so the answer is $\binom{12}{5} = 792$.

The general version: the number of k-element subsets of $\{1, \ldots, n\}$ in which any two elements differ by at least d is $\binom{n-(k-1)(d-1)}{k}$.

1.3 Recursion

When the objects have a natural size n, let a_n be the number of objects of size n and ask how an object of size n breaks up into a smaller object plus a last piece. Splitting by the last piece (or the first) gives an equation for a_n in terms of earlier terms. If the pieces you may add depend on what came before, one sequence is not enough: keep track of the possible endings as separate sequences, often called states.

Making a recursion rigorous.
A recurrence such as $a_n = a_{n-1} + a_{n-2}$ is a statement about a bijection: it says that the objects of size n split into two disjoint classes, one matched with the objects of size $n-1$ and the other with the objects of size $n-2$. In a proof, say what the classes are, why they do not overlap, why they cover everything, and what the matching is. Then check the first few values by hand.

Example 3.
How many strings of length 8 made from the letters A, B and C do not contain AB as two consecutive letters?

Solution.
One could count the strings with at least one AB and subtract from 3^8, but AB can occur in several places, and occurrences can be adjacent (as in ABAB), so inclusion-exclusion is laborious. A recursion is shorter.
(Recursion with states.) Whether we may add a B depends only on whether the string so far ends in A. So let x_n be the number of good strings of length n ending in A, let y_n be the number ending in B or C, and let $t_n = x_n + y_n$. A good string of length $n+1$ ending in A is any good string of length n followed by A, so $x_{n+1} = t_n$. A good string of length $n+1$ ending in B or C is either a good string ending in A followed by C, or a good string not ending in A followed by B or C, so $y_{n+1} = x_n + 2y_n$.
Then $t_{n+1} = t_n + x_n + 2y_n = 3t_n - x_n = 3t_n - t_{n-1}$, using $x_n = t_{n-1}$. With $t_1 = 3$ and $t_2 = 9 - 1 = 8$ we get

$$3,\ 8,\ 21,\ 55,\ 144,\ 377,\ 987,\ 2584,$$

so the answer is 2584.
These are alternate Fibonacci numbers: with $F_1 = F_2 = 1$, we have $t_n = F_{2n+2}$. This holds because $F_{m+2} = 3F_m - F_{m-2}$ for every $m \geq 3$, though the answer did not depend on noticing it.

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1.4 Combining the ideas

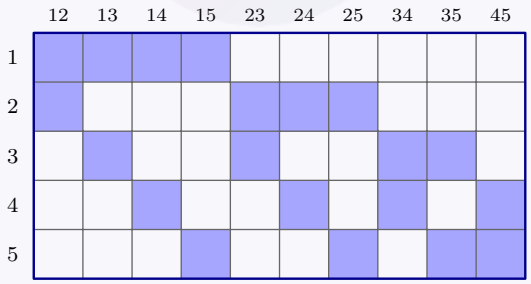
Harder problems mix the three. The next example uses double counting for a bound and a bijection for the construction; the first double count one tries gives a weaker bound than the second.

Example 4.
Some cells of a 5×10 grid (5 rows and 10 columns) are coloured so that no four coloured cells are the corners of a rectangle with sides along grid lines. What is the largest possible number of coloured cells?

Solution.
A rectangle is two rows and two columns with all four crossing cells coloured. So "no rectangle" says that two columns never share two coloured rows, or equivalently that two rows never share two coloured columns. Let N be the number of coloured cells.
First count pairs of coloured cells in the same row. If row i has r_i coloured cells, it uses $\binom{r_i}{2}$ pairs of columns, and no pair of columns may be used by two rows, so $\sum \binom{r_i}{2} \le \binom{10}{2} = 45$. With 24 cells in 5 rows the left side is at least $4\binom{5}{2} + \binom{4}{2} = 46$ (the most even split is $5, 5, 5, 5, 4$), so this gives $N \le 23$. That needs a smoothing argument to be rigorous, and the bound is weak in any case, since the answer is 20.
Now count pairs of coloured cells in the same column. If column j has c_j coloured cells, it uses $\binom{c_j}{2}$ pairs of rows, and there are only $\binom{5}{2} = 10$ pairs of rows to go round. So

$$\sum_{j=1}^{10} \binom{c_j}{2} \le 10.$$

Now $\binom{c}{2} - (c-1) = \frac{(c-1)(c-2)}{2} \ge 0$ for every integer $c \ge 0$, so $\binom{c_j}{2} \ge c_j - 1$. Adding over the ten columns gives $N - 10 \le 10$, so $N \le 20$.
The construction is fairly trivial once we look at the equality case. For equality every column must contain exactly two coloured cells and every pair of rows must be used exactly once. That is a bijection between the 10 columns and the 10 pairs of rows: in each column, colour the two rows of its pair. Two columns then share at most one coloured row, so there is no rectangle, and we have 20 coloured cells. The answer is 20.



The sharper bound came from counting pairs on the side with fewer possible pairs ($\binom{5}{2} = 10$ pairs of rows against $\binom{10}{2} = 45$ pairs of columns). When one double count is weak, try the other.

Common traps.

- Writing down a map and calling it a bijection without checking that it can be reversed. Give the inverse map explicitly, and check that each map lands in the right set.

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- Off-by-one errors in recursions. Check a_1 and a_2 (and a_0, if you use it) by hand before trusting the recurrence.
- Classes in a recursion that overlap or miss cases. Split by something definite, such as the last letter or the length of the last block.
- Ordered versus unordered pairs in double counting: a friendship is one unordered pair but two ordered ones.
- Fibonacci indexing. Say which convention you are using.

1.5 How to spot this in a contest

Phrases such as "each ... exactly k", "every two ... at most one" or "the sum over all subsets" point to double counting: count incidences or pairs, from both sides. If you need a bound, look for an inequality such as $\binom{c}{2} \geq c-1$, or an "as equal as possible" argument, to finish.

If small cases give binomial coefficients, powers of 2 or Fibonacci numbers, there is probably a bijection or a recursion behind them, so compute small cases first. Forbidden patterns ("no two adjacent", "no AB", "no three in a row") almost always mean a recursion with states. On the AIME the recurrence gets you a number; on the BMO you must also justify it, by stating the classes and the matching.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ In how many ways can a 1×12 strip be tiled with 1×1 squares and 1×3 rectangles?

Problem 2 ⋆⋆ For each non-empty subset of $\{1, 2, \ldots, 10\}$, write down its largest element. Find the sum of all the numbers written down.

Problem 3 ⋆⋆ How many strings of ten binary digits contain the block 01 (a 0 immediately followed by a 1) exactly three times?

Problem 4 ⋆⋆⋆ In a club, every member has exactly 5 friends among the other members (friendship is mutual). No two friends have a common friend, and any two members who are not friends have exactly 2 common friends. How many members does the club have?

Problem 5 ⋆⋆⋆ Let n be a positive integer. Call a subset of $\{1, 2, \ldots, 2n\}$ *balanced* if it contains as many odd numbers as even numbers (the empty set is balanced). Prove that the number of balanced subsets is $\binom{2n}{n}$.

Problem 6 ⋆⋆⋆⋆ In a group of nine people, every two are either friends or strangers. Call a set of three people *uniform* if the three are mutual friends or mutual strangers. Prove that there are at least 12 uniform sets.

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3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME I 2023 Problem 3** ⋆
   Count pairs of lines rather than points.
2. **BMO Round 1 2013/14 Problem 4** ⋆⋆
   Set up a recursion according to what happens on the last day, keeping track of states.
3. **AMC 10B 2019 Problem 25** ⋆⋆
   Find a recursion by looking at how a valid string can end.
4. **BMO Round 1 2023/24 Problem 1** ⋆⋆
   Ask which letters can end up in the last position, and recurse.
5. **BMO Round 1 2010/11 Problem 4** ⋆⋆
   Pair up arrangements so that the two in each pair have red counts of opposite parity, and see what is left unpaired.
6. **AIME I 2017 Problem 7** ⋆⋆⋆
   Read the sum as choosing a committee from three groups of six.
7. **BMO Round 1 2017/18 Problem 2** ⋆⋆⋆
   Count (swimmer, day) pairs in two ways, sorting the days by how many friends swim.
8. **AIME I 2018 Problem 10** ⋆⋆⋆
   Track the bug's position and set up a recursion, or count the two kinds of move.
9. **AIME I 2015 Problem 12** ⋆⋆⋆
   Add up the least elements by counting, for each k, the subsets whose least element is at least k.
10. **AIME I 2022 Problem 12** ⋆⋆⋆⋆
    Swap the order of summation: count element by element instead of pair by pair.
11. **BMO Round 1 2018/19 Problem 6** ⋆⋆⋆⋆
    Rotate the plane by 45 degrees so that each step changes both new coordinates by one.

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4 Hints

1. Look at the tile covering the rightmost cell.
2. Instead of going subset by subset, ask for each k how many subsets have largest element k, or how many have largest element at least j.
3. Put an extra 1 at the front and an extra 0 at the end. Now count the places where the digit changes.
4. Fix one member X and count the paths X, Y, Z in which Y is a friend of X and $Z \neq X$ is a friend of Y.
5. Keep the odd numbers of a balanced subset as they are, but swap its even part for its complement among the even numbers.
6. For a set of three that is not uniform, look at each of its members and ask whether that person is friends with exactly one of the other two. Count such "mixed corners" in two ways.

5 Solutions

Problem 1

In how many ways can a 1×12 *strip be tiled with* 1×1 *squares and* 1×3 *rectangles?*

Let a_n be the number of tilings of a $1 \times n$ strip. The rightmost cell is covered either by a square, leaving a $1 \times (n-1)$ strip, or by a 1×3 rectangle, leaving a $1 \times (n-3)$ strip. These cases do not overlap, so $a_n = a_{n-1} + a_{n-3}$ for $n \geq 3$, with $a_0 = a_1 = a_2 = 1$. Then

$$a_3, a_4, \ldots, a_{12} = 2,\ 3,\ 4,\ 6,\ 9,\ 13,\ 19,\ 28,\ 41,\ 60,$$

so there are 60 tilings.

Answer: 60

Problem 2

For each non-empty subset of $\{1, 2, \ldots, 10\}$*, write down its largest element. Find the sum of all the numbers written down.*

Write the largest element m of a subset as $m = \sum_{j=1}^{10} [m \geq j]$, where $[m \geq j]$ is 1 if $m \geq j$ and 0 otherwise. Swapping the order of summation, the required total is

$$\sum_{j=1}^{10} \#\{\text{non-empty subsets with largest element at least } j\}.$$

A subset has largest element less than j exactly when it is a subset of $\{1, \ldots, j-1\}$, so the number with largest element at least j is $2^{10} - 2^{j-1}$. The total is

$$\sum_{j=1}^{10} \left(2^{10} - 2^{j-1}\right) = 10 \cdot 1024 - (2^{10} - 1) = 10240 - 1023 = 9217.$$

(Alternatively, k is the largest element of 2^{k-1} subsets, and $\sum_{k=1}^{10} k\, 2^{k-1} = 9 \cdot 2^{10} + 1 = 9217$.)

Answer: 9217

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Problem 3

How many strings of ten binary digits contain the block 01 *(a* 0 *immediately followed by a* 1*) exactly three times?*

Given such a string s, form the 12-digit string s' by putting a 1 in front of s and a 0 after it. The new adjacent pairs at the ends read $1x$ and $x0$, so neither is 01, and s' contains 01 exactly as often as s does.

In s', look at the 11 gaps between consecutive digits and mark those where the digit changes. Since s' starts with 1 and ends with 0, the changes alternate $1 \to 0$, $0 \to 1$, $1 \to 0$, and so on, starting and finishing with a $1 \to 0$ change. So if there are three $0 \to 1$ changes there are four $1 \to 0$ changes: seven changes in all. Conversely, any choice of 7 of the 11 gaps, together with the first digit 1, determines a 12-digit string starting with 1, ending with 0 (an odd number of changes) and containing exactly three blocks 01; deleting its first and last digits gives a valid s. These maps undo each other, so the answer is $\binom{11}{7} = 330$.

Answer: 330

Problem 4

In a club, every member has exactly 5 *friends among the other members (friendship is mutual). No two friends have a common friend, and any two members who are not friends have exactly* 2 *common friends. How many members does the club have?*

Fix a member X and count the pairs (Y, Z) where Y is a friend of X, Z is a friend of Y, and $Z \neq X$. There are 5 choices of Y and then 4 choices of Z, so 20 pairs.

Count them by Z instead. Z cannot be a friend of X: otherwise the friends X and Y would have the common friend Z. So Z is one of the members other than X who are not friends of X, and each such member, having exactly 2 common friends with X, appears in exactly 2 pairs. Hence there are $20/2 = 10$ members who are neither X nor friends of X, and the club has $1 + 5 + 10 = 16$ members.

Such a club exists. Take as members the 16 strings of four binary digits, and make two strings friends when they differ in exactly one digit or in all four. Each string has 5 friends; one checks that friends have no common friend and that non-friends have exactly 2.

Answer: 16

Problem 5

Let n be a positive integer. Call a subset of $\{1, 2, \ldots, 2n\}$ balanced *if it contains as many odd numbers as even numbers (the empty set is balanced). Prove that the number of balanced subsets is $\binom{2n}{n}$.*

Let O and E be the sets of odd and even numbers in $\{1, \ldots, 2n\}$; each has n elements. For any subset X of $\{1, \ldots, 2n\}$ define

$$\varphi(X) = (X \cap O) \cup (E \setminus X),$$

that is, keep the odd elements of X and replace its even elements by the even numbers not in X.

Applying φ twice gives back X: the odd part is unchanged, and the even part is complemented twice. So φ is a bijection from the set of all subsets to itself, and it is its own inverse.

If X is balanced with k odd and k even elements, then $\varphi(X)$ has k odd elements and $n - k$ even elements, so it has exactly n elements. Conversely, if $|Y| = n$ with j odd elements, then Y has $n - j$ even elements, and $\varphi(Y)$ has j odd elements and $n - (n - j) = j$ even elements, so $\varphi(Y)$ is balanced. Therefore φ maps the balanced subsets into the n-element subsets and the n-element

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subsets into the balanced subsets, and since φ is its own inverse these two restrictions are inverse bijections. Hence the number of balanced subsets equals the number of n-element subsets of $\{1, \ldots, 2n\}$, which is $\binom{2n}{n}$.

(Counting balanced subsets directly by the value of k gives $\sum_{k=0}^{n} \binom{n}{k}^2$, so this also proves $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$.)

Problem 6

In a group of nine people, every two are either friends or strangers. Call a set of three people uniform *if the three are mutual friends or mutual strangers. Prove that there are at least* 12 *uniform sets.*

There are $\binom{9}{3} = 84$ sets of three people. Call a pair $(P, \{Q, R\})$, where P is a person and $\{Q, R\}$ is a pair of two other people, a *mixed corner* if P is friends with exactly one of Q and R. We count mixed corners in two ways.

By sets of three. In a uniform set $\{P, Q, R\}$ no corner is mixed. In a set that is not uniform, two of the three pairs are of one type and the third pair is of the other type. The person shared by the two pairs of the same type is not a mixed corner, and each of the other two people is joined to the others by one pair of each type, so is a mixed corner. So each non-uniform set contributes exactly 2 mixed corners, and if there are u uniform sets the number of mixed corners is $2(84 - u)$.

By people. If person P has d friends, then P has $8 - d$ strangers, and P is a mixed corner for exactly $d(8 - d)$ pairs. Since $d(8 - d) = 16 - (d - 4)^2 \leq 16$, the number of mixed corners is at most $9 \times 16 = 144$.

Hence $2(84 - u) \leq 144$, so $u \geq 84 - 72 = 12$.

(Equality is possible: arrange the nine people in a 3×3 array and make two people friends when they are in the same row or the same column. Everyone then has 4 friends, and the uniform sets are the 3 rows, the 3 columns and the 6 sets of three people in different rows and different columns.)

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